

Momentum Backtest - US Large Cap

Quantitative Investing / Asset Management Portfolio Project | 12-1 Momentum Strategy

10.1% Momentum CAGR	5.2% Benchmark CAGR	+4.9% Excess CAGR	61.9% Hit Rate
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Strategy Overview

The backtest evaluates a cross-sectional momentum strategy across a 20-stock US large-cap universe. At each monthly rebalance, stocks are ranked using cumulative returns from months t-12 through t-2. The most recent month is excluded to reduce exposure to short-term reversal effects. The strategy holds the four highest-ranked stocks at equal weights for one month, then rebalances.

Portfolio Construction

Component	Method
Universe	20 illustrative US large-cap equities
Momentum Signal	Cumulative total return from t-12 through t-2
Selection	Top 4 stocks / top 20% of universe
Weighting	Equal-weighted
Rebalance	Monthly
Transaction Costs	10 bps multiplied by portfolio turnover
Benchmark	Broad US large-cap market proxy

Backtest Results

Metric	Momentum	Benchmark
Annualized Return	10.1%	5.2%
Annualized Excess Return	+4.9%	-
Annualized Volatility	20.9%	Benchmark calculated in model
Maximum Drawdown	27.5%	Benchmark calculated in model
Monthly Hit Rate	61.9%	-

Investment Research Takeaways

- **Momentum effect:** the strategy systematically favors recent intermediate-term winners rather than relying on discretionary stock selection.
- **Implementation matters:** turnover is explicitly measured and transaction costs are deducted from gross strategy returns.
- **Risk-adjusted analysis:** CAGR is evaluated alongside volatility, Sharpe ratio, maximum drawdown and benchmark hit rate.
- **Signal discipline:** skipping the most recent month distinguishes the strategy from a simple trailing-12-month return screen.

Key Backtest Risks & Limitations

- **Survivorship bias:** a stable large-cap universe does not reproduce point-in-time index membership.
- **Market regime risk:** momentum can experience sharp reversals when market leadership changes quickly.
- **Implementation costs:** real bid-ask spreads, taxes, market impact and capacity constraints could reduce returns.
- **Data integrity:** a production backtest should use verified total-return data including corporate actions and delistings.

Recruiting / Interview Takeaway

I built a monthly US large-cap momentum backtest using a 12-1 signal. I ranked stocks by intermediate-term performance, selected the top quintile, equal-weighted the portfolio, incorporated turnover-based transaction costs and compared the strategy against a broad-market benchmark. I then evaluated CAGR, volatility, Sharpe ratio, drawdown and hit rate to determine whether the strategy generated meaningful risk-adjusted excess returns.

Methodology Note

The workbook uses an 84-month deterministic illustrative return series and therefore does not represent realized historical strategy performance. The methodology is based on established momentum research conventions, including the exclusion of the most recent month. For a production-quality study, the scenario inputs should be replaced with verified point-in-time total-return and constituent data. Educational and recruiting use only; not investment advice.